SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	418 I'm a Teapot
Selected Domain	Domain1
Selected SDG Track (1–17)	4 8 10 17
Project / Solution Name	AI Career Navigator
Name (Leader)	CHEN XIANGYU
Email & Phone (Leader)	CufoFlux@outlook.com +60173568532
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Artificial intelligence is rapidly transforming the labour market, creating new career opportunities while changing the skills required for existing jobs. However, workforce training and skills development programmes often struggle to keep pace with these changes. As a result, many individuals are uncertain about which skills they should develop to remain employable in an AI-driven economy.

This challenge particularly affects working adults whose jobs may be disrupted by AI, fresh graduates entering an increasingly competitive job market, and career switchers seeking new opportunities. Employers also face difficulties identifying future skills gaps and determining which training programmes will best prepare their teams for technological change.

In Malaysia, this issue is especially important as businesses and workers increasingly adopt AI tools, while access to relevant training remains a concern. Without timely and reliable career guidance, individuals may invest time and money in skills that become less relevant, while emerging opportunities in areas such as clean energy, healthcare, smart agriculture, and responsible AI may remain underexplored.

The problem has broader social and economic implications. Skills mismatches can contribute to unemployment, unequal access to career opportunities, and reduced productivity. At the same time, helping people transition into future-ready occupations can support lifelong learning, decent work, innovation, and sustainable economic growth.

Therefore, there is a need for an accessible and adaptive solution that helps individuals understand changing labour-market demands, identify relevant skills gaps, and make informed decisions about their future careers.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

AI Career Navigator is an AI-powered career guidance platform designed to help individuals prepare for changes in the future workforce. It transforms a user's current skills, experience, and interests into personalised career recommendations and actionable reskilling pathways.

Users begin by completing a skills profile through a simple intake interface. The system then analyses AI and labour-market trend signals from identifiable, updateable sources to identify emerging and declining skills. Based on the user's existing capabilities and the skills required by different future roles, the platform ranks suitable career options, explains each match score, and highlights the skills the user still needs to develop.

For each recommended role, AI Career Navigator provides a structured learning pathway with estimated learning time and accessible, free or low-cost learning resources. It also connects recommended careers and skills to relevant United Nations Sustainable Development Goals (SDGs), helping users understand how their career choices can contribute to sustainable development.

To account for uncertainty in AI adoption, the platform compares multiple future scenarios, such as moderate and rapid AI disruption. These scenarios help users explore alternative career pathways rather than treating predictions as certain outcomes.

The prototype uses a modular architecture that separates the user interface, backend API, career-matching logic, and data layer. Python supports backend processing and API development, while structured data and database storage support profile management and career recommendations. The team intends to build on relevant skills from the SDC Phase 1 workshops, including GenAI API integration, database architecture, and cloud hosting, where applicable.

AI Career Navigator is designed to support—not replace—human career judgement. Recommendations will be accompanied by sources and explanations, and predictions will be presented as estimates. The initial prototype will focus on demonstrating the connection between changing trend signals, explainable career matching, and personalised reskilling recommendations.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

The prototype is a responsive, build-free web application. The frontend uses HTML, CSS, native JavaScript ES modules, inline SVG, and browser `fetch`/`EventSource`. The Python 3 backend uses FastAPI and serves both the interface and JSON/SSE API from one origin. SQLite persists user profiles; browser `localStorage` caches interface state.

The backend loads versioned CSV tables covering Malaysian graduate labour statistics, MoHE graduate employment, international salary/employment comparisons, and source metadata. It combines these public indicators with an explicit skills taxonomy, role archetypes, learning-module catalogue, and three 2028 scenarios. A transparent weighted rules engine compares self-rated skills, experience, sector, and interests with role requirements. It returns role-fit scores and skill gaps, builds a time- and budget-aware reskilling pathway, and maps roles to the 17 UN SDGs. The narrative briefing is generated deterministically from these results and streamed over SSE. If the API is unavailable, the frontend can use a clearly labelled illustrative demo dataset.

SDC sponsor resources: Yes. The team’s technical declaration records use of SDC-provided Replit credits for development/deployment and hosting. No SDC sponsor-provided GenAI API is called by the current runtime. DeepSeek is listed in the team’s declaration as an AI-assisted development tool; the submitted backend currently uses its deterministic narrative engine rather than a live DeepSeek API call.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We began with the user challenge: graduates and workers need to choose a realistic career direction and understand what to do next as AI changes job requirements. We translated that need and the Domain 1 requirements into one connected journey: capture a skills profile and constraints, compare future roles with visible reasons and evidence, then turn a selected role into a practical learning pathway. The four-step intake keeps the effort manageable; dated sources, explicit scoring factors, and scenario comparisons make recommendations inspectable rather than opaque. We separated the frontend, API contract, data catalogue, and scoring engine so the prototype can be extended without rebuilding the user journey. For Phase 2, deploy the same-origin app on Replit, replace the placeholder demo URL, and pilot the full flow with target users. Use feedback to refine intake wording and navigation; validate mobile accessibility, source freshness, API failure recovery, privacy handling, and pathway usefulness before a wider release.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 4.4 (primary): substantially increase the number of youth and adults with relevant skills for employment, decent jobs and entrepreneurship. Contributing targets: 8.5 (transition into decent work) and 10.2 (equitable access to career guidance and reskilling).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	35.5% of employed degree and diploma holders in Malaysia — approximately 1.96 million people — were in skills-related underemployment in Q3 2025, working in low- or semi-skilled jobs below their qualification level (DOSM).
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Core Measurable KPI: Skills-gap closure rate — the percentage of identified priority skill gaps that are closed. Definition: For each user, the share of skill gaps identified for their selected target role at intake whose post-pathway reassessment level meets or exceeds the role-required level within 90 days. Cohort value = total priority gaps closed ÷ total priority gaps identified × 100. Example: A user has 5 priority gaps at intake and closes 2 by reassessment; the user's skills-gap closure rate is 40%.
Phase 2 Impact Target	Achieve a 30% skills-gap closure rate across the Phase 2 pilot cohort by the final prototype presentation — at least 30% of the priority skill gaps identified at intake are closed,

Metric Category	Your Response
Your realistic, percentage-based target by the final prototype presentation *e.g., "20% improvement in computational resource use."*	with post-pathway reassessment showing that the user meets the selected role's required skill level.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes √ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

√ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
FastAPI	code	MIT License
Pydantic	code	MIT License
Openpyxl	code	MIT License
Uvicorn	code	BSD 3-Clause license
Pandas	code	BSD 3-Clause License
DeepSeek-V4.1-Flash-1017	model	MIT License
Malaysia Graduates Statistics 2024	dataset	DOSM — Government Open Data Terms of Use 1.0
Malaysia Graduates Statistics 2023	dataset	DOSM — Government Open Data Terms of Use 1.0
Malaysia Graduates Statistics 2022	dataset	DOSM — Government Open Data Terms of Use 1.0
Malaysia Graduates Statistics 2021	dataset	DOSM — Government Open Data Terms of Use 1.0
Malaysia Graduates Statistics 2020	dataset	DOSM — Government Open Data Terms of Use 1.0
Malaysia Graduates Statistics 2019	dataset	DOSM — Government Open Data Terms of Use 1.0
Malaysia Graduates Statistics 2018	dataset	DOSM — Government Open Data Terms of Use 1.0
Malaysia Graduates Statistics 2024 infographic PDF	dataset	DOSM — Government Open Data Terms of Use 1.0
Malaysia Graduates Statistics release Archive	dataset	DOSM — Government Open Data Terms of Use 1.0

Malaysia Publication / 数据申请入口	dataset	DOSM — Government Open Data Terms of Use 1.0
Malaysia Graduate Employment Rate portal	dataset	MoHE/KPT — None
Malaysia Graduate Employability portal	dataset	MoHE/KPT — None
United States Winter 2026 Salary Survey Executive Summary	dataset	NACE — None
United States 2026 Student Survey	dataset	NACE — None
United States Labor market outcomes by college major 2026	dataset	Federal Reserve Bank of New York via CBS/La Nación — None
United Kingdom Graduate Outcomes 2023/24	dataset	HESA — Creative Commons 4.0 CC-BY
United Kingdom Drop in graduate employment 'a cooling, not a collapse'	dataset	Times Higher Education — None
Germany Akademikerarbeitslosigkeit 2025	dataset	CC BY-NC-ND
Germany Handwerk oder Studium	dataset	CC BY-SA 4.0; CC BY-NC-ND 4.0 DE
Japan 令和 7 年度大学等卒業者の就職状況調査	dataset	CC BY 4.0
Japan 2026 届应届生起薪 25.4228 万日元/月	dataset	None
South Korea 2024 年毕业生就业统计	dataset	None
South Korea OECD Education at a Glance 2026 via UPI：25–64 岁大学毕业生就业率 80.1%（2025	dataset	UPI — None
Australia 2025 Graduate Outcomes Survey National Report	dataset	QILT — None
Australia 2025 GOS-L (三年后中位薪资 A$91,000)	dataset	QILT — None
Canada Labour market outcomes of postsecondary graduates	dataset	Statistics Canada — Statistics Canada Open Licence
Canada Characteristics and median employment income by educational qualification and field of study	dataset	Open Government Portal — Open Government Licence – Canada

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes √ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] 陈相宇 CHEN XIANGYU

[Date] 25/10/2026